LPPS-IKHC: Location Privacy-Preserving Scheme Using Improved k-Anonymity and Hybrid Cache for IoV

Yufeng Li, Bo Wang, Qi Liu, Xiangyu Zheng, Jiangtao Li, Yiwei Wang, Jiajun Xi, and Wutao Qin

***Abstract*—As the Internet of Vehicles (IoV) evolves, integrating Location-Based Services (LBS) into in-vehicle software has become increasingly common. Consequently, protecting users' location privacy has emerged as a critical concern. Vehicles face the challenge of sending query requests to untrusted Location Service Providers (LSP) through untrusted Roadside Units (RSU) to obtain relevant results. This situation presents two primary risks: 1) the involvement of untrusted RSU increases the likelihood of monitoring vehicle communications with the LSP, and 2) repeated queries to the untrusted LSP increase the risk of privacy breaches during interactions. This paper proposes an innovative location privacy protection scheme for continuous LBS in the IoV, called LPPS-IKHC, which combines improved k-anonymity with hybrid cache. The scheme utilizes anonymous techniques and hybrid cache to mitigate the risk of exposing sensitive information to untrusted entities. To protect against adversaries, LPPS-IKHC generates specific anonymous sets in the presence of untrusted RSU. Additionally, an anonymous algorithm incorporating trajectory similarity is introduced to protect trajectory privacy, further enhancing user location privacy. Furthermore, by reducing the number of interactions between vehicles and untrusted LSP, hybrid caching helps diminish the risk of privacy disclosure. Extensive experimental results demonstrate that the LPPS-IKHC method provides superior privacy protection and lower system overhead compared to existing schemes.**

***Index Terms*—Hybrid cache, Internet of Vehicles, location privacy protection, k-anonymity.**

I. Introduction

With the rapid development of the Internet of Vehicles (IoV) and advancements in information technology, intelligent connected vehicles are becoming increasingly integrated into daily life. According to the International Energy Agency (IEA), global electric vehicle sales reached nearly 14 million units in 2023, reflecting a 35% increase compared to 2022 [1]. By leveraging vehicle communication terminals, information processing systems, and wireless technology, the IoV enables a wide range of location-based services for users.

However, the widespread adoption of IoV technologies has introduced significant challenges, with location privacy protection emerging as a critical concern [2]. Navigation systems highlight this issue, as they depend on precise location data to provide users with optimal routes, real-time updates, and traffic forecasts. While these services deliver substantial benefits, they also pose privacy risks. Malicious actors can exploit location data to deduce sensitive personal information, such as home and work addresses, behavioral patterns, and daily routines, thereby compromising user privacy [3]. Consequently, safeguarding vehicle users' location privacy has become a pressing issue requiring immediate attention and practical solutions.

Existing approaches to privacy preservation in Location-Based Services (LBS) can be broadly categorized into four types: anonymization-based mechanisms, differential privacy mechanisms, cryptographic-based mechanisms, and cache-based mechanisms.

Anonymization-based mechanisms protect user locations by obscuring them within a specified area or among multiple locations, preventing adversaries from obtaining precise information. Techniques such as k -anonymity [4], and its extensions like l-diversity [5] and t-closeness [6], introduce $k-1$ decoy locations to form an anonymity set, reducing the probability of identifying the user's real location to $1/k$. Differential privacy mechanisms add noise to location data to prevent personal identification, introducing randomness during data publishing or query processes. This ensures that even with access to a dataset or query results, an individual cannot discern the precise location of a specific user [7].

Cryptographic-based mechanisms safeguard sensitive data's confidentiality and integrity using advanced encryption algorithms [8], secure multi-party computation [9], spatial transformation [10], and private information retrieval techniques [11]. Cache-based mechanisms utilize caching technologies to store and manage location data, reducing interactions with untrusted Location Service Providers (LSP) and mitigating the risk of privacy breaches [12].

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However, these methods face inherent limitations and fail to address the two risk factors identified in the abstract. First, anonymization-based mechanisms do not account for vehicle networking background information [13], [14] and are vulnerable to replay attacks from untrusted Roadside Units (RSU), which can infer the user's real location [15]. Additionally, repeated queries still require interactions with untrusted LSP, increasing the risk of privacy breaches. Second, differential privacy mechanisms degrade data quality, availability, and accuracy due to the noise added for privacy preservation [16], [17], [18]. This noise can obscure precise location data for LBS, reducing service accuracy and leading to potentially incorrect recommendations. Third, cryptographic-based mechanisms, while offering provable security under specific models, impose high computational overhead, making them impractical for resource-constrained IoV devices [19], [20]. Finally, existing cache-based mechanisms inefficiently utilize cached data for future requests and require frequent communication with untrusted LSP, increasing communication costs and the risk of privacy breaches [21], [22], [23].

To overcome these limitations, we propose a location privacy protection scheme for IoV, termed LPPS-IKHC, which combines improved k-anonymity and hybrid cache. The LPPS-IKHC leverages vehicle network background information to generate anonymous sets, providing robust protection against untrusted RSU. To further enhance user location privacy, the scheme incorporates trajectory similarity into the anonymization algorithm, ensuring effective trajectory privacy protection. Additionally, hybrid caching is employed to minimize interactions between vehicles and untrusted LSP, significantly reducing privacy risks and communication costs. The hybrid caching algorithm improves data availability by optimizing the cache hit ratio through considerations of data freshness and contribution, thereby enhancing overall efficiency.

The main contributions of this article are as follows:

1) We propose an improved k-anonymity mechanism integrated with hybrid cache to mitigate the risk of sensitive information exposure to untrusted entities in IoV.
2) The study devises a novel k-anonymity-based privacy mechanism that utilizes IoV contextual information and incorporates trajectory similarity for enhanced location privacy protection.
3) Our approach develops a hybrid cache strategy combining active and passive techniques, reducing vehicle requests to untrusted LSP and minimizing unnecessary data exposure.
4) The research conducts a comprehensive security analysis and experimental evaluation, demonstrating that LPPS-IKHC achieves superior privacy protection and lower system overhead compared to existing approaches.

The paper structure encompasses several sections that contribute to a comprehensive analysis. Section II reviews the state-of-the-art in related work. The problem statement is presented in Section III, along with the system model and fundamental concepts essential to this study. Section IV outlines our primary recommendations, the core of this work. Section V focuses on performance evaluation and in-depth analysis to shed light on the practical aspects of our recommendations. Concluding the paper, Section VII summarizes our findings and proposes future research directions.

II. Related Work

Existing location privacy-preserving mechanisms can be classified into four main categories: anonymization-based, differential privacy, cryptographic-based, and cache-based.

A. Anonymization-Based Mechanisms

Anonymization-based mechanisms safeguard personal privacy by obscuring individual identities and concealing sensitive data. These techniques employ location perturbation and obfuscation methods while integrating established privacy frameworks. A widely used approach is k-anonymity [31], which ensures that a user's true location is indistinguishable from $k-1$ other locations, reducing an attacker's probability of accurately identifying the true location to $1/k$. However, traditional k-anonymity is vulnerable to attacks exploiting background knowledge [32]. To address these limitations, advanced models such as l-diversity [5] and t-closeness [6] have been proposed. l-Diversity extends k-anonymity by ensuring that each anonymized group contains at least l well-represented values for sensitive attributes, thereby improving diversity. t-Closeness further refines l-diversity by ensuring that the distribution of sensitive attributes in each equivalence class closely matches their overall distribution in the dataset. Despite these advancements, centralized location anonymizers [33] present a potential Single Point of Failure (SPOF). Alternative schemes, including cloaking [28], dummy location generation [31], [34], [35], and anonymization techniques [23], [36], [37], have been developed to distort user information. However, these approaches often fail to account for adversaries' access to secondary information, such as query probabilities based on location and time or semantically relevant data associated with queries.

Moreover, traditional anonymization schemes primarily focus on obscuring the user's real location without leveraging cached query results for future requests. This oversight increases the risk of sensitive information leakage and results in unnecessary communication overhead caused by redundant interactions with the LSP.

B. Differential Privacy Mechanisms

Differential privacy protects individual location privacy by adding noise or perturbations to publicly accessible location data, rendering it impossible to accurately determine an individual's real location. Its robust mathematical foundation has made differential privacy a widely favored approach among researchers. Yang et al. [16] proposed a data publishing mechanism using group perception technology to achieve differential privacy, ensuring robust location protection by accounting for user density and the non-uniform distribution of data during partitioning. Li et al. [38] identified the challenge of determining effective anonymous regions due to the noise introduced by differential privacy. To address limitations in existing LBS systems, Zhang et al. [39] developed an algorithm that protects both

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user location and query privacy by considering spatiotemporal correlation and background knowledge. Their method employs the Voronoi diagram for map partitioning, selects false location points using an improved k-anonymity algorithm combined with l-diversity, and applies the Laplacian mechanism to enhance location privacy.

While effective, differential privacy introduces noise into the data, potentially reducing its accuracy and quality. This trade-off can impact the reliability of data analysis and applications.

C. Cryptographic-Based Mechanisms

Cryptographic-based mechanisms for location privacy protection encrypt users' LBS query information to ensure that, even if a query request is intercepted or monitored during transmission, the user's true location cannot be derived from the encrypted ciphertext. These approaches effectively fulfill location privacy protection requirements by providing robust security and privacy assurances. Yi et al. [40] utilized Paillier homomorphic encryption to achieve differential location privacy and geographic discriminability. Paulet et al. [41] proposed a nearest-neighbor search method leveraging Oblivious Transfer (OT) [42] and Private Information Retrieval (PIR) [27] to retrieve LBS query information while maintaining high query accuracy. Chen et al. [43] introduced Circular Shifting Encryption Protocols (CSEP), which use homomorphic encryption to encode user locations, enabling LSP to compute distances directly from ciphertext. Jannati et al. [44] employed an ElGamal-based homomorphic encryption scheme, allowing clients to encrypt location coordinates securely. Deng et al. [45] designed a location privacy protection scheme using random encryption cycles.

However, it is important to recognize that encryption methods, while enhancing location privacy, impose significant computational overhead. This added complexity can make such approaches impractical for mobile devices with limited computational resources.

D. Cache-Based Mechanisms

Cache-based mechanisms have emerged as practical solutions to mitigate the risk of sensitive information leakage by reducing interactions with untrusted entities [23], [46], [47], [48]. By allowing users to access desired content from caches, these mechanisms eliminate the need to send queries to untrusted LSP. Niu et al. [29] proposed a method leveraging base station caching to strengthen location privacy protection. Their approach quantified cache usage and location privacy exposure, introduced entropy-based metrics, and optimized cache hit rates by considering factors such as data freshness. Chen et al. [12] developed a method combining active caching and k-anonymous co-optimization to further enhance location privacy. Albuk et al. [21] introduced a double obfuscation mechanism that integrates two obfuscation methods with a fog caching system to improve location privacy protection. Furthermore, Zhang et al. [15] proposed a trajectory privacy-preserving scheme that utilizes a transition matrix and caching techniques to safeguard location privacy in continuous LBS.

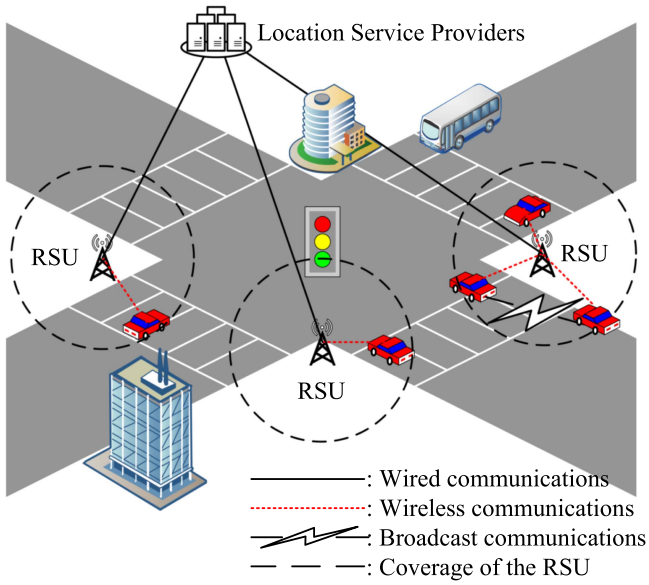


Fig. 1. System model.

Despite their advantages, traditional cache-based mechanisms exhibit limitations. They predominantly focus on protecting static location information, often overlooking privacy concerns related to user trajectories. This gap can expose users to privacy breaches, as movement patterns may be inferred from cached queries or subsequent LBS requests.

E. Critical Review

Table I provides a comprehensive summary of location privacy technologies. In the threat model, "RSU" and "LSP" denote the capability to address challenges associated with untrusted RSU and LSP, respectively. "Repetition" and " Trajectory" indicate the system's ability to manage repeated queries and safeguard trajectory privacy. The proposed LPPS-IKHC scheme incorporates improved k-anonymity and hybrid cache techniques to effectively address privacy risks arising from untrusted RSU and repeated location queries in the IoV. Additionally, the scheme addresses the limitations of existing methods, which often lack comprehensive evaluations of privacy protection, efficiency, and communication overhead.

III. Problem Formulation

In this section, we present the system model of LPPS-IKHC, followed by its design goals and Attacker model. Before providing detailed descriptions, we define the notations used throughout the paper in Table II.

A. System Model

The system model, depicted in Fig. 1, consists of three key entities: the LSP, RSU, and vehicles. When a vehicle requests LBS, the process unfolds as follows:

1) *Step 1:* The vehicle transmits its current location to the RSU.

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TABLE I
COMPARATIVE ANALYSIS OF LOCATION PRIVACY-PRESERVING TECHNIQUES

Methods	Technology	Threat model			Description
		RSU LSP	Repetition	Trajectory	
Anonymization-based	k-anonymity [24] l-diversity [5] t-closeness [6]	✓	✗	✗	The anonymity-based mechanism adds communication overhead and neglects the protection of trajectory privacy.
Differential privacy	Laplace [16] Gaussian [17] Staircase [18]	N/A	✗	✗	The noise added by differential privacy can reduce theaccuracy and quality of data.
Cryptographic-based	PHEP [25] OT [26] PIR [27]	✓	✗	✓	Limited computing power in IoV devices hinders the effectiveness of encryption mechanisms.
Cache-based	DLS [28] enhanced-DLS [29] Sky-Cache [30]	✓	✓	✗	Existing caching-based mechanisms have limitations in efficiently utilizing caching for future requests.
LPPS-IKHC	Improved k-anonymity and Hybrid Cache	✓	✓	✓	Utilizing improved k-anonymity and hybrid caching to effectively address privacy risks from untrusted RSU and repeated location queries in the IoV.

TABLE II
SUMMARY OF IMPORTANT NOTATIONS

Notation	Description
k	k of initial and expanding anonymous set
H	Entropy of anonymous set
T	Lifetime of cached data
F	Data freshness
D	Fréchet distance
R	Real trajectory
F	Virtual trajectory
Tr	Trajectory formed
t	Time since the cell data was cached
q_i	Probability the ith location is query location
p_i	Query frequency
L_i	Request for the ith LBS service of the anonymous set
C_j	The jth group's set
λ	Average uncertainty
γ	Cache hit ratio

2) *Step 2:* The RSU checks its cache. If the required data is found, the RSU will directly respond to the vehicle. Otherwise, the RSU forwards the request to the LSP.
3) *Step 3:* The LSP provides the necessary information to the RSU, updates the RSU's cache, and the RSU delivers the data to the vehicle.

The primary roles and assumptions of each entity are detailed below:

- *Vehicle:* Vehicles are equipped with onboard units featuring limited computational and storage capacities. These units include various sensors, a Tamper-Proof Device (TPD), and communication modules. Vehicles commonly utilize Cellular Vehicle-to-Everything (C-V2X) [49] or Dedicated Short-Range Communication (DSRC) protocols for interaction with other infrastructure or vehicles. In this study, vehicles generate virtual locations to form anonymous sets based on their real locations, thereby enhancing privacy and security (Step 1).
- *RSU:* The RSU is deployed along roadways and employ DSRC technology [50] to communicate with vehicles' onboard units (OBU). Their primary function is to cache Point of Interest (POI) data and serve as intermediaries for vehicle communication with the LSP (Step 2).
- *LSP:* The LSP handles vehicle requests and delivers updated POI data [51] to vehicles via the RSU. Based on the location and keywords provided by vehicles, the LSP supplies relevant POI information (Step 3).

B. Attacker Model

This study addresses the presence of untrusted RSU and LSP, focusing on internal threats as the primary attack mode. These threats are considered actions initiated by the LSP and RSU.

In the threat model, the LSP is assumed to monitor anonymized queries across the system, enabling access to all location query data. Simultaneously, the RSU can unlawfully intercept query information by eavesdropping on user communications. Moreover, both the LSP and RSU are presumed to possess full knowledge of the location privacy-preserving mechanisms implemented in the system.

Based on the available information, untrusted RSU and LSP can execute inference attacks, replay attacks [52], and Viterbi attacks [53] to compromise user location privacy. The inference attack involves deducing whether a specific location is a valid query location using the adversary's available knowledge. The replay attack occurs when an adversary identifies a location in the cloaking set as the real query location, leveraging system knowledge to re-run the location privacy-preserving mechanisms and determine if the generated cloaking set corresponds to the real query location. The Viterbi attack is a trajectory-based attack. The attacker utilizes the Viterbi algorithm to decode and recover the most likely true trajectory by exploiting the spatiotemporal correlation in the user's location trajectory, thereby compromising location privacy.

C. Design Goal

The system model delineates the roles of entities within the LPPS-IKHC framework and describes their operational processes. The threat model identifies two potential adversaries: the RSU, which may eavesdrop on communications between vehicles and the LSP, and the LSP, whose repeated interactions with vehicles exacerbate privacy risks. To mitigate these threats, this study introduces a location privacy protection scheme for continuous LBS in the IoV. The proposed scheme integrates

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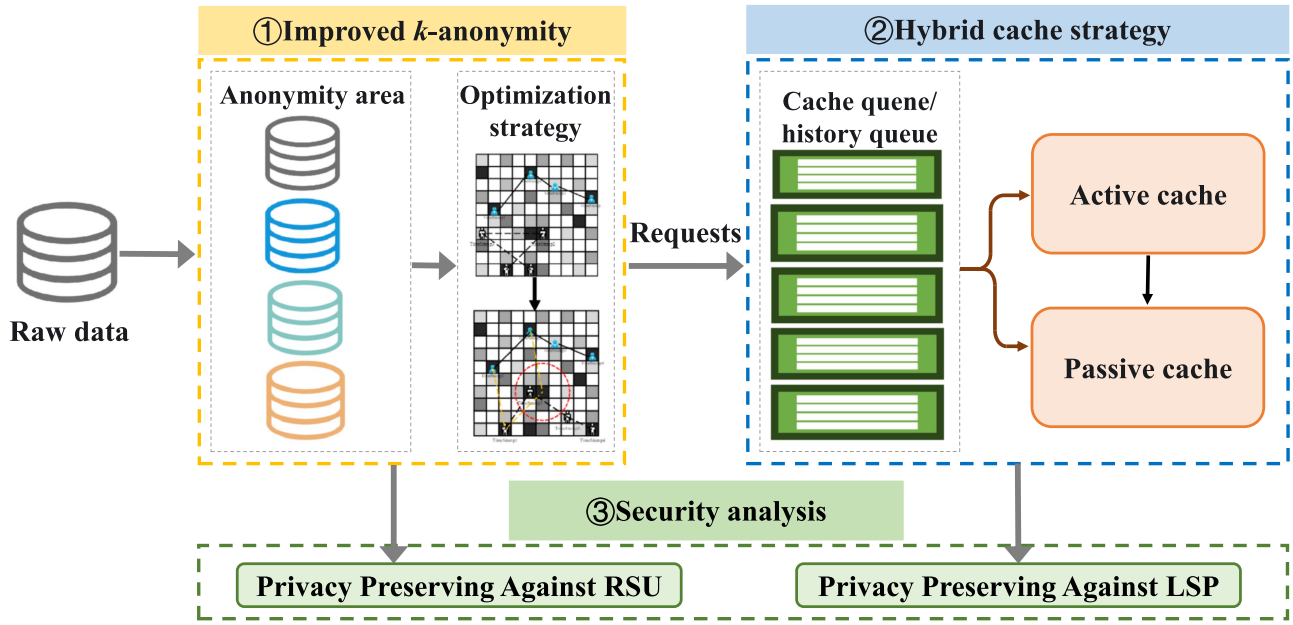


Fig. 2. The workflow of LPPS-IKHC.

improved k-anonymity and hybrid cache techniques to reduce the exposure of sensitive data to untrusted entities. Specifically, LPPS-IKHC generates anonymized sets in environments with untrusted RSU and incorporates trajectory similarity measures to protect trajectory privacy. Additionally, hybrid caching minimizes interactions with untrusted LSP, further lowering privacy risks. For more details, refer to Section IV of this paper.

IV. Proposed LPPS-IKHC Scheme

In this section, we present the workflow of the proposed location privacy protection scheme. First, we outline the construction of k-anonymity and the design of the hybrid cache strategy. Finally, we conduct a security analysis to demonstrate the effectiveness of the proposed scheme in preserving location privacy.

A. Overview of LPPS-IKHC

This study introduces a robust and efficient location privacy protection scheme tailored for the IoV. The core concept leverages an RSU as a caching node to store POI data.

As illustrated in Fig. 2, the vehicle employs an improved k-anonymity algorithm to generate k locations, which it then sends to the RSU. Upon receiving a request, the RSU checks its cache for matching data. If a match is found, the RSU directly responds to the vehicle with cached data, avoiding LSP involvement. This approach minimizes the vehicle's location disclosure and reduces communication overhead. If no match exists, the cache is updated using the Least Recently Used-2 (LRU-2) algorithm to enhance the cache hit rate.

Additionally, our study includes a comprehensive security analysis that examines potential attack vectors involving untrusted RSU and LSP. We demonstrate how the LPPS-IKHC scheme effectively mitigates these threats, ensuring location privacy and preventing data breaches.

B. Improved k-Anonymity

This section introduces an enhanced fake location generation algorithm designed to address the limitations of conventional k-anonymity algorithms in the context of the IoV. Although the traditional k-anonymity algorithm provides a baseline level of location privacy, it is less effective against sophisticated attack methods. To overcome these shortcomings, we build on the traditional algorithm to develop a more robust and secure strategy for generating fake locations.

Our approach involves a comprehensive analysis of the deficiencies in traditional k-anonymity algorithms, particularly in preserving location trajectory privacy within intelligent connected vehicle applications. These identified issues are systematically addressed in the algorithm design. The optimized fake location generation strategy retains the advantages of the original k-anonymity algorithm while significantly enhancing its privacy protection capabilities.

1) k-Anonymity: k-anonymity is the most extensively studied anonymity model, primarily due to its intuitive anonymization process. In simple terms, a dataset D satisfies k-anonymity if each Quasi-Identifier (QI) value, denoted as $D(QI)$, appears in at least k records. The fundamental goal of k-anonymity is to thwart record linkage by ensuring that an individual remains indistinguishable within ak-anonymous group. This approach balances privacy protection and information preservation, aiming to minimize the distortion required to conceal an individual within the group. It is important to note that achieving optimal k-anonymity has been proven to be an NP-hard problem [54].

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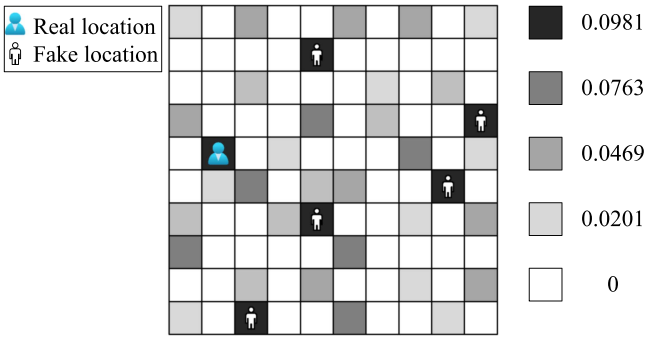


Fig. 3. Distribution of anonymous location sets for $k = 6$.

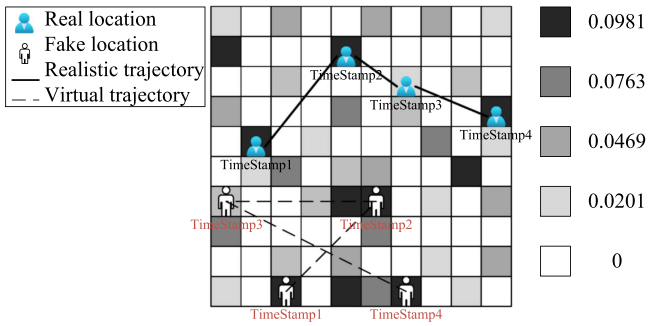


Fig. 4. Comparison of true and false trajectories when $k = 2$.

In this study, the algorithm focuses on generating a set of plausible virtual locations. We begin by dividing the map area into a grid of $n \times m$ equally sized cells, as illustrated in Fig. 3. Each cell is assigned a query probability derived from historical service requests, which are analyzed based on recent data from Google Maps. The query probability is expressed in (1):

$$q_i = \frac{c_i}{P_m}, i = 1, 2, 3, \ldots, m \times n; \text{ where } \sum_{i=1}^{m \times n} q_i = 1. \quad (1)$$

Here, c_i denotes the probability of queries occurring within cell i, and P_m represents the probability of queries across the entire map.

The proposed algorithm searches a large database to identify the optimal set of fictitious locations. To satisfy the anonymity requirement of k, in addition to the real location, $k-1$ additional cells are selected as virtual locations. The algorithm addresses this challenge as follows:

- *Selecting an Anonymity Level:* Users specify an appropriate anonymity level, k, which balances location privacy and computational overhead. A higher k value provides stronger anonymity but increases system overhead due to the complexity of selecting virtual locations.
- *Maximizing Privacy through Entropy:* When all k locations have an equal probability of being the true location, the system achieves maximum entropy, indicating the highest level of privacy. The algorithm begins by accessing the query probabilities for all positions in the RSU. It then ranks all cells by the absolute difference between their query probability and the true position's probability. From this ranking, the algorithm selects the $4k$ cells with the smallest differences (this set size is adjustable). Users then generate m groups, each containing k cells, where one cell is the true location, and the remaining $k-1$ cells are randomly selected from the candidate set. The jth $(j \in [1, m])$ group is expressed as $C_j = [c_{j_1}, c_{j_2}, \ldots, c_{j_i}, \ldots, c_{j_k}]$. Based on the original query probabilities of the selected cells, the normalized query probabilities of the included cells can be denoted as $p_{j_1}, p_{j_2}, \ldots, p_{j_i}, \ldots, p_{j_k}$, calculated using (2).

$$p_{jk} = \frac{q_{ji}}{\sum_{l=1}^{k} q_{jl}}, \quad i = 1, 2, 3, \ldots, k. \quad (2)$$

The algorithm aims to identify an optimal set of anonymous locations that achieves efficient k-anonymity. The proposed solution leverages entropy as a metric to ensure the desired privacy level. Entropy, a widely accepted measure of user privacy, is maximized for a given set C_j, as defined in (3).

$$H_j = -\sum_{i=1}^{k} pj_i * \log_2 pj_i. \quad (3)$$

Here, pj_i represents the normalized result of the cells in Cj.

2) Optimization Strategy: This section examines the challenges associated with applying k-anonymity to vehicular networks and presents the proposed solutions. Niu et al. [28] suggest sorting the probabilities of all positions within the RSU coverage and selecting the position with the closest probability to the real location as one of the $k-1$ fictitious positions. However, this method often results in tightly clustered locations within the k-position set, which can compromise privacy by inadvertently disclosing the real location. To address this, Niu introduced the concept of the Cloaking Region (CR), designed to maximize the spatial distance between locations and obscure the true position more effectively.

In the vehicular networking context, rigorous location filtering may leave too few positions for effective privacy protection. This limitation can enable attackers to infer a user's trajectory through edge information as shown in Fig. 4. To mitigate this, our paper proposes an enhanced strategy.

Improving the privacy of the DLS algorithm involves optimizing both entropy and trajectory similarity. This dual objective is framed as a Multi-objective Optimization Problem (MOP) [55]. One objective is to maximize entropy-based privacy protection, while the other is to enhance trajectory privacy by increasing the similarity between virtual and real trajectories.

How should we measure the similarity of trajectories? In this paper, we employ the Fréchet distance [56] as our metric for trajectory similarity. Let (S, d) be a metric space, where d represents a metric function on S. In our context, the real trajectory R and the virtual trajectory F are continuous curves defined on S. The Fréchet distance between R and F is defined as follows:

$$D(R, F) = \max\{d(R(t), F(t)\}. \quad (4)$$

Here, d denotes the metric function on S, and in this paper, we define d as CR, specifically, $d(x, y) = CR(x, y)$. Let $C =$

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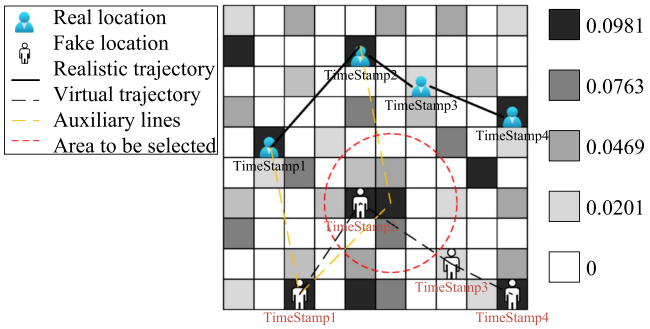


Fig. 5. The main method to protect the privacy of real location trajectory.

$[c_1, c_2, \ldots, c_k]$ represent the set of current real and virtual positions, and let $Tr' = [tr_1, tr_2, \ldots, tr_k]$ represent the trajectory formed by the set of real and virtual positions at the previous time as well as the current real and virtual positions. We can formulate this MOP as follows:

$$\max \left\{ -\sum_{i=1}^{k} p_i * \log_2 p_i, -\sum_{i=1}^{k} D(tr_i, tr_{Real}) \right\}. \quad (5)$$

Here, tr_{Real} represents the user's real trajectory, maximizing the inverse of the Fréchet distance aims to maximize trajectory similarity.

In this MOP, we face the challenge of simultaneously maximizing both functions. To address this, we select $4k$ positions (a user-defined parameter) with the smallest absolute probability differences as candidates, thereby satisfying the entropy requirement. Regarding the location trajectory issue in vehicular networks, when a vehicle transmits an anonymous location set, it retains a record of the previously sent location set. This sequential data reveals a movement trend known as the trajectory. Leveraging this information, we select false locations with similar trajectories and alter them to enhance trajectory privacy protection. As depicted in Fig. 5, the real location trajectory, evolving over time, is mapped to virtual locations, and the search for false locations is expanded within the region. Achieving similarity in trajectories contributes to more robust location privacy.

Finally, the "optimal" k-anonymous set A is output by Algorithm 1.

Algorithm 1: k-Anonymity Optimization.

Require: The real location of the vehicle L_r and the anonymous collection from the previous time step K_l
Ensure: An anonymous collection of locations generated from the true vehicle location
1: $All \leftarrow$ Location_probability() ▷ Obtain the probabilistic mapping of all cells in the RSU range
2: $KA \leftarrow \emptyset$ ▷ Initialize the set of KA and use it to store the final result
3: **for** i in All **do**
4: $All \leftarrow$ Sort_by(i, L_r) ▷ Sort the cell-to-probability mapping in ascending order by $|i - L_r|$
5: $KA \leftarrow$ All$[1 : 4k]$ ▷ Select the first 4k positions after sorting as candidates for KA
6: **for** Ka in KA **do**
7: $Max\{-\sum_{i=1}^{k} p_i * \log_2 p_i, -\sum_{i=1}^{k} D(tr_i, tr_{Real})\}$ ▷ Select the position with higher trajectory similarity and feasibility using the formula
8: **end for**
9: **end for**
10: **return** $\max\{KA[k-1] \cup \{L_r\}\}$ ▷ Return the result

C. Hybrid Cache Strategy

The privacy metric defined in (3) quantifies users' privacy when sending queries to the LSP but does not account for the impact of caching on overall privacy. Caching enhances privacy by reducing the necessity of sending certain queries to the LSP. To address this, this paper introduces an additional privacy metric: the Global Privacy Metric.

When considering all queries in the system, the uncertainty of the real location for queries answered by the LSP is calculated using H_j in (3). However, the LSP remains uninformed about the user's real location for queries answered by the cache, leaving each cell as a potential real location. leaving each cell as a potential real location. Consequently, the uncertainty of the true position is defined as $\log_2(N^2)$, assuming an equally likely distribution among all cells.

The second privacy metric is defined as the average uncertainty (λ) of the true location across all queries, where H_q represents the uncertainty of the real location in query q, calculated using (7). This metric can be simplified concerning the cache hit ratio, as the reduced equation shows.

$$\lambda = \frac{\sum_{q \in Q_{server}} H_q + \log_2 N^2 * |Q_{cache}|}{|Q_{cache}| + |Q_{server}|}. \quad (6)$$

$$\gamma = \frac{|Q_{cache}|}{|Q_{cache}| + |Q_{server}|}. \quad (7)$$

$$\lambda = \frac{\sum_{q \in Q_{server}} H_q}{|Q_{cache}| + |Q_{server}|} + \gamma * \log_2 N^2. \quad (8)$$

There are two evident ways to increase λ: one is by increasing the entropy of each query $q_{Q_{server}}$, and the other is by raising the cache hit ratio γ, allowing more users to access the cache directly and benefit from higher privacy protection. The system considers both of these factors to safeguard location privacy.

Consider a scenario with a single RSU connected to remote service providers. Given that the cache capacity of the RSU is limited, each RSU has a specific cache capacity denoted as C. The area within each RSU's range is subdivided into square units. Using recent data from Google Maps, the probability of service requests in each square location area within the RSU, denoted as p_i, can be determined. In the Internet of Vehicles, where vehicle users typically request similar types of LBS services, it is assumed that each LBS service cache occupies an equivalent portion of the RSU's cache, facilitating subsequent hybrid cache updates.

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Algorithm 2: Active Cache.

Require: The positions S of all cells in the RSU range and their corresponding probabilities P, along with the capacity V of the RSU
Ensure: Actively cache the optimal policy selected
1: $F \leftarrow \emptyset$ ▷ Initialize the cache policy
2: **for** i in range(0, V) **do**
3: $F \leftarrow F \cup \{S[\text{get_max_index}(P)]\}$ ▷ Find the index of the maximum probability and add the corresponding position to the result
4: $S \leftarrow S - S[\text{get_max_index}(P)]$ ▷ Remove the selected position from the set of available positions
5: $P \leftarrow P - P[\text{get_max_index}(P)]$ ▷ Remove the corresponding probability
6: **end for**
7: **return** F

Algorithm 3: Passive Cache.

Require: The vehicle requests RSU to send a k-anonymous set of size k
Ensure: Updated RSU cache policy
1: Queue $queueF \leftarrow \emptyset$ ▷ Initialize the history queue to store request information using the FIFO rule
2: Queue $queueC \leftarrow$ Active cache ▷ Initialize the cache policy
3: **for** i in k **do**
4: **while** i.size + $queueF$.size $\geq C1$ **do** ▷ Limit the size of the history queue
5: $queueF$.pop()
6: **end while**
7: $queueF$.push(i) ▷ Update the history queue
8: **if** $queueF$.same() == true **then**
9: **while** same.size + $queueC$.size $\geq C2$ **do**
10: $queueC$.pop()
11: **end while**
12: $queueC$.push(same) ▷ Insert the repeatedly accessed service into the cache queue
13: **end if**
14: **end for**
15: **return** $queueC$

1) Active Cache: The vehicle obtains services through either the RSU or the LSP. When the RSU initiates a service request to the LSP for a cache update, it must transmit the vehicle's location information to the LSP. In this paper, we refer to this process as active caching, which essentially initializes the RSU's cache.

It is well understood that the RSU's cache size is limited, which presents the challenge of optimizing cache utilization and enhancing the cache hit rate. Building on previous research, which assumed that requests for the same service yield results of equal size, the critical question becomes: how can we maximize the cache hit rate when the number of buffers is fixed, thereby maximizing the cache probability?

To achieve this objective, we employ a greedy algorithm during the cache initialization phase. Specifically, as shown in Algorithm 2, the algorithm selects services that contribute the most to the cache—meaning those with the highest probabilities within the designated size.

2) Passive Cache: This article establishes two queues in the RSU: a cache queue and a history queue. In the context of vehicle networking, requests for LBS services from vehicles are represented by specific codes, so we assume these requests do not occupy cache space.

The cache queue stores LBS service query results, while the history queue records request information. As shown in Algorithm 3, When an LBS service is initially requested, the relevant information is added to the history queue. If the requested data does not receive at least two visits while in the history queue, it is removed according to the First In First Out (FIFO) rule.

When data in the history queue has been accessed twice, it is removed from the history queue, placed in the cache queue, and cached. The cache queue is then sorted by time. When a cached LBS service in the data queue is reaccessed, it is moved to the front of the cache queue. Data elimination prioritizes older data at the end of the cache queue, particularly data since the second-to-last visit.

Furthermore, to maintain data freshness, if the cache has not been updated for a significant period, or if the freshness reaches a certain threshold, the RSU automatically updates this portion of the cache and positions it at the front of the cache queue.

Cache contamination occurs when certain data is accessed infrequently, sometimes only once. If this data remains in the cache, it consumes valuable cache space. To address this, we employ the LRU-2 algorithm in this paper, which reduces cache pollution issues and improves the hit rate compared to the standard Least Recently Used (LRU) algorithm.

When a user sends k location requests, the system first checks the cache pool. If the required information is found, there is no need to send a request to the LSP, thereby reducing the risk of privacy disclosure. If the relevant information is not in the cache pool, a request is made to the cloud for the service, and the cache is updated according to the LRU-2 algorithm. Additionally, we consider cache freshness, and when freshness is too low, the RSU automatically requests an update for the corresponding service from the LSP.

Fig. 6 illustrates an example of the hybrid cache strategy in an application scenario with $k = 3$ and three vehicles within the coverage area of a single RSU. The hybrid cache architecture utilizes anonymous location data (L_i) generated by users through the k-anonymity algorithm. Using an active caching policy, the RSU stores location service data from the LSP in its cache. The cache framework depicted in Fig. 6 includes the Cache Queue (which consumes storage space) and the Request Queue (which does not consume storage space).

The RSU initializes its cache using a greedy algorithm. Vehicles send requests sequentially from right to left. When CarA sends its request, the queue is updated based on the FIFO rule, and the cache responds promptly.

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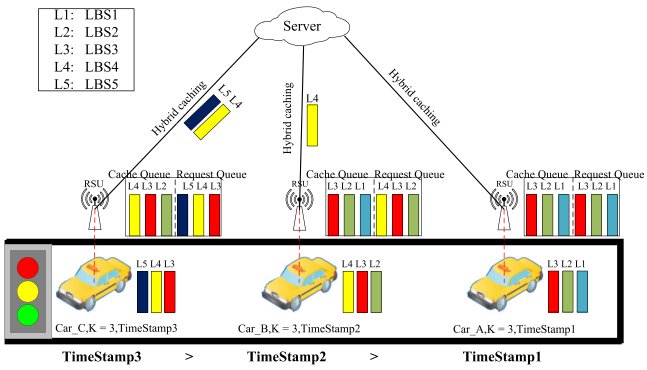


Fig. 6. An example of hybrid caching.

In the second instance, CarB's request updates the queue and identifies cache hits for L_2 and L_3. The RSU then requests information on L_4 from the cloud and relays it to the vehicle.

During the third iteration, CarC's request updates the queue, caches L_3, and places it at the forefront. The RSU requests information on L_4 and L_5 from the cloud. Adjustments are made to the cache queue since L_4 was part of the previous request's queue. Specifically, the L_1 service at the tail is removed, and L_4 information is inserted at the head, completing the cache update. This strategy enhances location privacy and improves system performance.

D. Security Analysis

In this section, we conduct a comprehensive security analysis of the proposed LPPS-IKHC scheme to demonstrate how it effectively enhances user location privacy. Leveraging the threat model described in Section III-B, we analyze the security of the LPPS-IKHC framework and show how it can resist the threats posed by untrusted RSU and untrusted LSP.

1) Privacy Preserving Against LSP: In instances where the cached data stored in the RSU satisfies the users' requirements, they can directly access the relevant query results without involvement from the LSP. This feature allows users to obtain the desired information while ensuring complete security and privacy, as the LSP cannot access any details about the query.

However, when the cache fails to provide satisfactory query results, the RSU forwards the query to the untrusted LSP. In the context of the LPPS-IKHC scheme, where the LSP is considered an active adversary, we identify and analyze two distinct attack surfaces that pose potential threats: $inference\ attack$ and $replay\ attack$ [52].

Theorem 1: The LPPS-IKHC scheme is designed to be resilient against replay attacks for untrusted LSP.

Proof: In this context, we consider the scenario where the attacker has obtained the anonymization algorithm through reverse engineering or other means. The attacker then attempts to leverage the knowledge of the algorithm to guess the real location of user U out of the submitted k locations. Specifically, the adversary ascertains the user's real location by inputting all locations within the anonymous set into the anonymization algorithm and subsequently comparing the newly generated anonymous set to the original one to identify any discrepancies.

However, since (3) ensures high uncertainty for each location, the attacker is unable to pinpoint the real location, even with an understanding of how the LPPS-IKHC algorithms work. Furthermore, the design of the LPPS-IKHC algorithm involves randomly selecting $2k$ positions from $4k$ and then optimizing them, introducing a certain level of randomness. Additionally, the presence of the RSU cache ensures that the LSP acquires all anonymous locations, minimizing the probability of a successful replay attack by an adversary.

Theorem 2: The LPPS-IKHC scheme is designed to be resilient against inference attacks for untrusted LSP.

Proof: According to the attack model described in Section III-B, an LSP can be considered an active adversary. By compromising the LBS server and accessing information from all users, the LSP gains global information and can perform inference attacks. The attacker knows the query probabilities for the entire map, including historical and current queries. This information encompasses the user's identifier, a mix of real and dummy locations, interests, query range, and other relevant data. Based on such information, the adversary can conduct inference attacks to obtain sensitive details about the user. Specifically, the active adversary has access to the following data: the query probability q_i for each group cell C, the user's interest I, and all submitted locations $l_1, l_2, \ldots, l_c, \ldots, l_k$. Let $P_G(event)$ represent the probability that the adversary can accurately determine the truth of the $event$.

Definition 1: A scheme is inference attack resistant [28] if

$$p_G\{l_i \in U | U \in C\} = p_G\{l_i \in U | U \in C\}, \quad \forall (0 < i \neq j \leq k). \tag{9}$$

For each submitted location l_i in the proposed LPPS-IKHC, the probability of a successful guess can be computed as follows:

$$p_G\{l_i \in U | U \in C\} = \frac{p_G\{l_i \in U \cap U \in C\}}{p_G\{U \in C\}} = \frac{q_j}{p_G\{U \in C\}}. \tag{10}$$

Similarly, for the submitted location l_j, the probability of a successful guess is:

$$p_G\{l_j \in U | U \in C\} = \frac{q_j}{p_G\{U \in C\}}. \tag{11}$$

Thus, for the pair of submitted locations l_i and l_j, (9) holds if

$$q_i = q_j, \forall i \neq j. \tag{12}$$

In our schemes, according to (3), maximum entropy is achieved when all potential dummy locations have the same probability, ensuring that (12) is satisfied.

2) Privacy Preserving Against RSU: When a user initiates an LBS request, the vehicle device generates an anonymous set containing k distinct pieces of information, such as the query identifier and the queried point of interest. Given this limited information, the probability that the RSU can determine the real query location is only $1/k$. However, if the RSU gains access to the entire anonymous set, we must consider an extreme scenario where the RSU and LSP collude to launch attacks. In such cases, the RSU can perform $replay\ attacks$ and $Viterbi\ attacks$ [53].

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Theorem 3: The LPPS-IKHC scheme is designed to be resilient against replay attacks for untrusted RSU.

Proof: Similar to Theorem 1, in this scenario, the RSU has access not only to the LPPS-IKHC algorithm but also to the entire anonymous set, enabling it to launch replay attacks. In addition, (3) ensures a high level of uncertainty for each location, rendering the attacker incapable of pinpointing the real location, even with a thorough understanding of the LPPS-IKHC algorithms. Moreover, the LSPPS-IKHC algorithm is designed to randomly select $2k$ positions from $4k$ and then optimize them, thereby introducing a significant degree of randomness.

Theorem 4: The LPPS-IKHC scheme is designed to be resilient against viterbi attacks for untrusted RSU.

Proof: To compromise trajectory privacy, the RSU can employ the Viterbi attack [15]. This method involves analyzing a set of anonymous groups created by the same user to infer real movement trajectories. Our approach utilizes the Fréchet distance to measure the similarity between trajectories [53], [57]. By emphasizing trajectory similarities, we aim to make it more challenging to identify individual users based on their movement patterns. Specifically, we use (3) and (5) to select virtual locations with more substantial rule similarity to incorporate into the anonymous set, thereby enhancing the difficulty for adversaries to identify individual users and ensuring anonymous protection of trajectory privacy.

V. Evaluation

In this section, we present a detailed empirical evaluation to highlight the advantages of the proposed privacy protection scheme. We start by describing the datasets, evaluation metrics, and comparison methods used in our experiments. Subsequently, we provide a comprehensive report and analysis of the experimental results.

A. Experiment Setting

Datasets: Various types of trajectory datasets have been utilized to assess the effectiveness of existing privacy models. These datasets include sources such as taxi trips, user check-ins, phone call records, and Bluetooth readings, among others. For our study, we have selected the publicly accessible SNAP-Gowalla dataset [58] to systematically compare the privacy protection models discussed in Section IV. The SNAP-Gowalla dataset consists of user location check-ins recorded from February 2009 to October 2010 and includes 196,591 nodes and 950,327 edges. Additionally, we have generated several synthetic datasets based on SNAP-Gowalla, each with unique characteristics such as dataset size, mean track length, and data location compliance. These synthetic datasets will facilitate a more comprehensive evaluation of our privacy protection model.

Experimental Environment: The experiments in this study were conducted on a system running Ubuntu 24.10 x86_64 with kernel version 6.11.0-9-generic. The hardware configuration includes an Intel Xeon Silver 4314 processor with 64 cores, operating at 3.4 GHz, paired with an ASPEED Technology ASPEED GPU (Model 02:00.0). The system is equipped with 12 GB of memory, providing ample computational resources for the simulation tasks. The software environment was configured with Python 3.12.7, utilizing key scientific libraries such as NumPy, SciPy, and Matplotlib to support the simulations and data visualizations.

Evaluation Metrics: We compare privacy models using various performance criteria, including privacy metrics, utility metrics, and computational costs. Based on a summary of existing evaluation indicators, we select several representative privacy and utility indicators for our analysis.

- The information entropy of an anonymous set quantifies the uncertainty in distinguishing between real and virtual locations within the set. Assume that the anonymous set AS contains one real vehicle location, with the remaining locations being virtual and generated based on the real one. For example, in a k-anonymity scenario, the k location information from k query requests sent to the RSU forms an anonymous set. We represent the real location as r and the virtual locations as d. The information entropy of the anonymous set is defined as follows:

$$H = -\sum_{d \in AS} p(r,d)\log_2(r,d). \tag{13}$$

  Here, $p(r,d)$ represents the probability of correctly distinguishing between the real position r and the virtual position d. A higher entropy value indicates that the LSP has less information about the true vehicle location and is less likely to deduce it. To ensure effective location privacy protection, this paper considers the uncertainty between real and virtual locations. This uncertainty is quantified using information entropy as a metric, which provides a more accurate evaluation of the scheme's performance in location privacy protection.
  Simultaneously, the cache hit rate plays a crucial role in determining the level of location privacy protection. A higher cache hit rate means that vehicle users can access services through the cache more frequently without needing to send requests to the LSP, which effectively reduces the risk of privacy disclosure.
  Therefore, the location privacy protection measure proposed in this paper comprehensively considers both the information entropy and the cache hit ratio of the anonymous set. This metric provides a thorough evaluation of the scheme's effectiveness in protecting location privacy. Adopting this holistic approach gives a clearer understanding of the proposed scheme, highlighting ways to enhance it for more efficient and reliable location privacy protection.
- The cache hit ratio is a crucial metric that measures the proportion of successfully retrieved queries from the RSU cache relative to the total number of queries made by the vehicle. A cache hit indicates that the sought-after POI data is available within the RSU cache. The cache hit ratio is defined as follows:

$$\gamma = \frac{|Q_{cache}|}{|Q_{all}|}. \tag{14}$$

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Here, Q_{all} represents the POI query requests initiated by the vehicle, while Q_{cache} denotes the successful query requests found in the cache. A higher cache hit ratio indicates a reduction in the number of queries the vehicle sends to the LSP, thereby enhancing the vehicle's location privacy protection. The cache hit ratio reflects the efficiency of the solution in minimizing communications with the LSP. A higher cache hit ratio means that the vehicle can more frequently access the required POI data from the RSU cache, thereby decreasing the frequency of communication with the LSP and reducing the risk of compromising location privacy.

This paper provides a comprehensive evaluation of the location privacy protection scheme by considering both information entropy and the cache hit ratio of the anonymous set. When combined, these two metrics offer a holistic assessment of the scheme's effectiveness in safeguarding location privacy. This approach enables readers to understand the proposed scheme thoroughly and provides insights for enhancing its efficiency and reliability in protecting location privacy.

- To provide a clearer representation of the overall level of privacy protection, including the effects of caching, we introduce a third privacy metric: the average uncertainty λ regarding the true location across all queries:

$$\lambda = \frac{\sum_{q \in Q_{server}} H_q}{|Q_{cache}| + |Q_{server}|} + \gamma * \log_2 N^2. \tag{15}$$

Here, H_q represents the uncertainty of the real location in query q, Q_{cache} denotes the set of queries served by the cache, and Q_{server} signifies the set of queries sent to and served by the LSP. The cache hit ratio, denoted by γ, fluctuates dynamically through the hybrid cache strategy. The inclusion of $log_2 \tilde{} N^2$ highlights the impact of the anonymization set size (N) on the overall privacy measurement, representing the privacy metric without considering the cache and defining the uncertainty of the true position with an equally likely distribution among all cells.

Comparing experiment: We will compare and analyze the results with the CaDSA location privacy protection scheme [29], the DLS algorithm [28], and Hema et al.'s k-anonymity and cache scheme for Unmanned Aerial Vehicles (UAV) [30]. To comprehensively evaluate these location privacy protection models, we will assess their entropy (H), cache hit ratio (γ), average uncertainty (λ), time overhead, and communication overhead across different privacy protection levels (k) and various time intervals.

B. Entropy

In this study, we experimentally evaluated the relationship between the k value and the level of location privacy protection, as quantified by entropy. Fig. 7 illustrates the performance of various schemes in terms of privacy protection, specifically in relation to entropy values. Generally, as the k value increases, entropy also tends to increase. To provide a clearer representation of the entropy data across different scenarios, we use the "Optimal" scenario to denote the maximum attainable entropy ($\log_2 k$) under the given conditions.

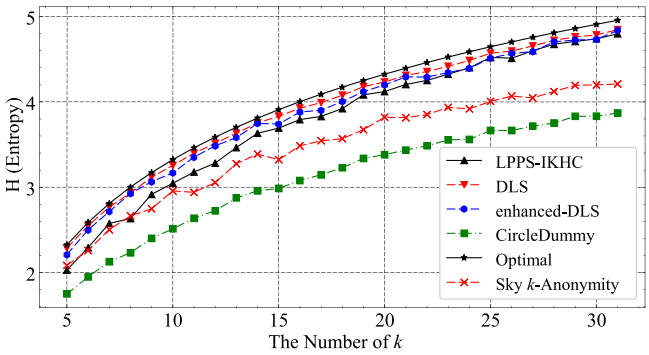


Fig. 7. Comparison of entropy across different methods with respect to k.

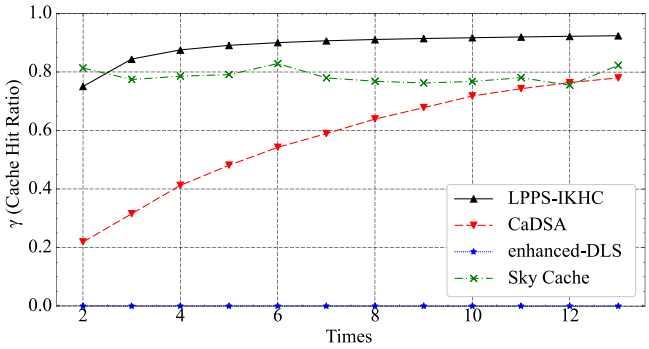


Fig. 8. Comparison of cache hit ratio across different methods over time.

The "CircleDummy" scheme consistently restricts the selected region to a virtual circle, leading to less diverse location choices and, therefore, lower entropy values. In contrast, both the DLS scheme and the approach proposed in this paper demonstrate significant improvements in privacy protection, achieving higher entropy levels. This enhancement is attributed to the selection of virtual locations from cells with similar query probabilities, which ensures greater entropy.

Compared to the "UAV" scheme, our proposed method demonstrates lower entropy. This decrease can be attributed to the gradual changes in the user's anonymity range as they move, which complicates selecting the optimal anonymity scheme. Additionally, our method shows a slight reduction in entropy compared to the DLS scheme. It is because, while aiming to maximize entropy, we also account for the similarity of location trajectories. By incorporating this consideration, we trade a portion of the entropy for enhanced trajectory privacy protection.

C. Cache Hit Ratio

1) Cache Hit Ratio vs. Time: Fig. 8 illustrates the evolution of the cache hit ratio over time T. In our experiment, we set $k = 10$, which represents the size of the generated anonymous set. Our focus is on cache updates, and we expect the cache hit ratio to gradually increase and eventually stabilize at a higher value over time.

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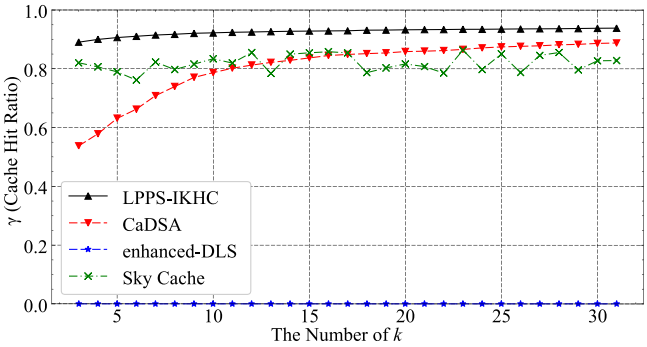


Fig. 9. Comparison of cache hit ratio across different methods with respect to k.

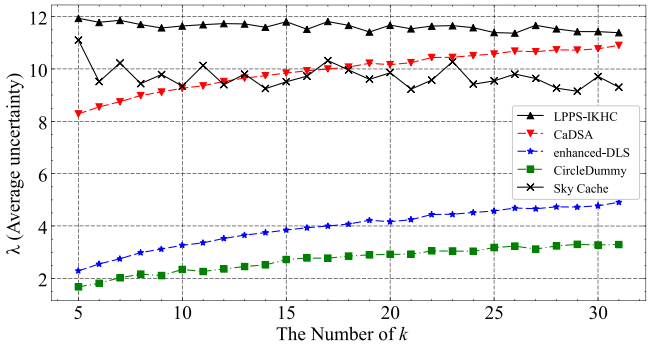


Fig. 10. Comparison of λ and k across different methods.

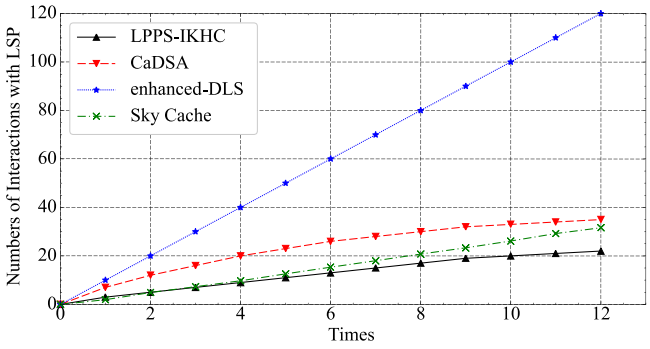


Fig. 11. Comparison of request numbers between different methods over time.

Our method initially does not perform as well as the UAV method because it generates randomness in the anonymous set to resist replay attacks, leading to a lower initial cache hit rate.

However, compared to CaDSA and the enhanced DLS, our proposed scheme demonstrates a superior cache hit rate during the RSU's active cache phase. Subsequently, with the application of the LRU-2 algorithm for updates, the cache hit ratio steadily increases until it reaches a stable state. In contrast, the UAV scheme uses the LRU algorithm as its caching strategy, resulting in significant fluctuations in the cache hit rate and ultimately yielding inferior performance compared to our approach.

2) Cache Hit Ratio vs k: As the value of k increases, the size of the anonymous set sent each time also increases. However, an imperfect cache configuration can lead to more requests being sent to the LSP, increasing the communication overhead for the user. Fig. 9 illustrates that the cache hit ratio for the enhanced DLS algorithm remains consistently low, effectively zero, due to the lack of cache implementation.

In contrast, both CaDSA and the algorithm proposed in this paper incorporate a caching mechanism, gradually improving the cache hit rate as k increases. Notably, the proposed algorithm utilizes the LRU-2 caching strategy and considers privacy factors such as data freshness, providing an advantage over CaDSA.

However, inherent randomness in the cache location selection and anonymization generation can cause CaDSA to outperform LPPS-IKHC, particularly when k exceeds 65. It aligns with other studies that typically set k between 0 and 30, considering values beyond 65 impractical for our context [15], [23], [28], [59], [60].

In contrast, the SkyCache algorithm, which uses the LRU approach, struggles to efficiently update the cache when dealing with large-scale request data. It leads to significant fluctuations in the cache hit rate. Consequently, the proposed algorithm shows better performance in this context.

D. The Average Uncertainty

To effectively illustrate the comprehensive impact of caching on privacy protection, this paper examines how the average uncertainty (λ) of real locations across various algorithms changes with different values of k. This relationship is depicted in Fig. 10.

It is evident that the Enhanced DLS algorithm provides the least privacy protection, as it does not incorporate caching. In contrast, the approach presented in this paper significantly enhances overall privacy protection compared to both CaDSA and SkyCache.

A comparison of the average uncertainty (λ) shown in Fig. 10 with the entropy values in Fig. 7 reveals that while the proposed strategy sacrifices some entropy, it substantially improves overall privacy protection.

E. Communication Overhead

Users can efficiently access content via RSU communication without incurring additional costs [12], [61]. However, if content retrieval through the RSU fails, users must resort to the cellular network, which incurs communication fees.

In typical scenarios, users interact with service providers via the RSU, resulting in communication fees. Fig. 11 illustrates the temporal evolution of missed cache requests over time (t). Notably, in our scenario, the communication overhead increases at a slower rate, reflecting more efficient caching strategies and a reduced reliance on the cellular network. This improved efficiency leads to lower overall communication costs for users.

Additionally, the frequency of interactions with the LSP provides a more practical and representative measure of communication costs. Each interaction with the LSP usually incurs a fee, making it a direct indicator of the cost burden compared to merely counting the number of requests. By quantifying these interactions, we can better evaluate communication overhead

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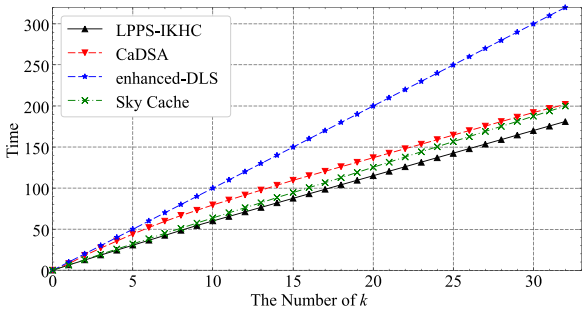


Fig. 12. Comparison of LBS service time across different methods as k increases.

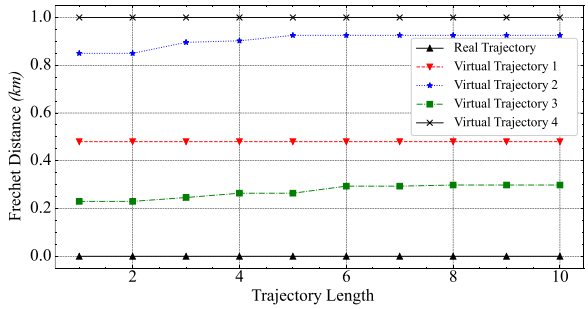


Fig. 13. Changes in trajectory similarity with increasing trajectory length.

and privacy risks, enabling a more comprehensive understanding of communication costs.

F. Time Cost

The time cost associated with varying anonymity levels (k) is evaluated in Fig. 12. In this study, time cost is defined as the total execution time, where each instruction's duration is treated as a single unit. This definition is applied in a controlled simulation environment, enabling theoretical comparisons by quantifying time cost as the product of the total number of executed instructions and their assumed execution duration. As depicted in the figure, the time cost for all schemes increases with higher k values. Specifically, the proposed LPPS-IKHC scheme is evaluated against the average time costs of the CaDSA, enhanced-DLS, and SkyCache schemes across various k levels.

Fig. 12 demonstrates that the time cost for these schemes follows a linear growth trend as k increases from 5 to 30. This growth is primarily driven by the expansion of the stealth set, which demands greater computational effort to achieve higher anonymity levels. The simulation also considers the overhead introduced by RSU cache switching. In particular, when a request is made, the RSU cache update is incorporated into the instruction tree, ensuring this factor is accounted for in the overall time cost calculation.

Notably, the LPPS-IKHC scheme consistently outperforms the other three schemes in terms of time cost. This advantage is primarily attributed to its efficient caching mechanism, implemented through the LRU-2 algorithm, which minimizes redundant computations and reduces communication overhead. By leveraging this algorithm, the LPPS-IKHC scheme effectively mitigates the additional computational burden associated with increasing k values, maintaining superior efficiency across varying anonymity levels.

G. Trajectory Similarity

To evaluate the effectiveness of trajectory protection, we conducted an experiment utilizing the Fréchet distance to measure trajectory similarity. The trajectory length and the Fréchet distance were plotted on the horizontal and vertical axes, respectively, to visualize the impact of the proposed trajectory privacy protection method. For the experiment, the RSU coverage radius was set to 1000 meters.

In the experimental setup, we configured $k = 5$, with one real trajectory and four virtual trajectories. The Fréchet distance for the real trajectory was set to zero. As shown in Fig. 5 and Fig. 13, the primary objective was to minimize the increase in the Fréchet distance to preserve trajectory similarity. In other words, the aim was to maintain the Fréchet distance as low as possible, ensuring a high degree of similarity between real and virtual trajectories. Trajectories 1 and 4 represent ideal cases where the Fréchet distance remains unchanged, indicating that optimal virtual positions were successfully selected to maximize entropy. In contrast, Trajectories 2 and 3 were unable to identify suitable virtual positions within the constraints of the Fréchet distance. Consequently, these trajectories increased the Fréchet distance (represented by the radius of the circle in Fig. 5) to find more appropriate positions.

VI. Discussion

This study presents an innovative location privacy protection scheme for continuous LBS in the IoV, referred to as LPPS-IKHC. The proposed scheme combines anonymity techniques with a hybrid caching approach to reduce the risk of sensitive information exposure to untrusted entities, such as RSU and LSP. Simulation results demonstrate significant improvements across multiple key performance metrics.

A. Advantages of LPPS-IKHC

Location privacy protection in the IoV has gained considerable attention in recent years. However, this study acknowledges that even when a vehicle's location is obfuscated with noise, attackers may still infer its trajectory. This inference is possible because attackers can exclude certain virtual points by exploiting background information from the vehicle network. To address this issue, the model proposed in this paper incorporates trajectory privacy protection. The following sections detail the novelty of our approach, emphasizing the proposed model, hybrid caching technique, and its advantages over existing solutions.

- *Threat Model and Privacy Protection:* In this study, we define the active attacker as the LSP and the passive attacker as the RSU, allowing for a clearer distinction of their roles and capabilities within the vehicular network. We examine several attack modes that can threaten the privacy of vehicular trajectories, including replay attacks, inference attacks, and Viterbi attacks, with a primary focus on

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protecting trajectory privacy, a critical element of location-based services in the IoV. A detailed security analysis of the proposed method demonstrates its effectiveness in addressing these threats and ensuring trajectory privacy.

- *Key Innovations of the Approach:* This work introduces several novel techniques to enhance location privacy in the IoV. First, we leverage background information from the IoV to optimize the selection of virtual locations, effectively obscuring the vehicle's true trajectory. Additionally, we incorporate trajectory similarity to constrain location selection at time $t + 1$, ensuring that virtual locations align with the vehicle's movement patterns, thereby further protecting privacy. To improve data retrieval efficiency, we implement a hybrid caching mechanism: a greedy algorithm initially populates the cache to achieve a high hit rate, while the LRU-2 algorithm updates the cache over time, improving stability and maintaining a high cache hit rate. These innovations work together to strike an effective balance between privacy protection and system performance in the IoV.
- *Performance and Efficiency:* Our method outperforms existing schemes in both privacy protection and system overhead. In terms of privacy, the global average privacy measure (λ) of our approach exceeds that of current methods. Additionally, our scheme reduces communication overhead, leading to lower traffic costs. It also demonstrates improved time efficiency, requiring less processing time compared to existing solutions.

B. Limitations

While the proposed scheme demonstrates strong theoretical results, several limitations should be considered:

- *Dynamic Nature of Wireless Channels and Vehicle Movement:* We primarily focused on theoretical analysis within a controlled simulation environment, the inherent randomness introduced by the dynamic nature of wireless channels and vehicle movement in real-world vehicular networks can affect the reliability and efficiency of information transmission. These factors may lead to performance variations that were not captured in the simulations conducted. Despite these limitations, the proposed approach demonstrates strong theoretical potential and feasibility under the considered conditions. Future work will address these challenges by integrating more realistic models of wireless channel fluctuations and vehicle dynamics, which will enhance the robustness and applicability of the proposed scheme in real-world scenarios.
- *Definition and Measurement of Time Cost:* In this study, we define time cost based on the execution time of individual instructions, assuming each instruction takes a unit of time. While this approach simplifies the analysis and is suitable for theoretical exploration, it does not fully account for the complexities of real-world time costs. Specifically, the influence of vehicle motion and wireless channel variability on time cost was not considered in our experiments. In future research, we plan to incorporate more detailed models of time cost that account for these variables, providing a more accurate representation of time efficiency in practical scenarios.

VII. Conclusion

We propose a location privacy-preserving scheme that combines enhanced k-anonymity with a hybrid caching approach for the IoV. The scheme uses the LRU-2 algorithm to optimize caching, reducing interactions with the LSP and minimizing privacy risks. By efficiently managing data access, it helps obscure vehicle locations while maintaining privacy.

Security analysis confirms the scheme's effectiveness in protecting location privacy, demonstrating resilience against attacks like replay and Viterbi attacks. Experimental results show significant reductions in system overhead while improving privacy protection.

In future work, we aim to integrate machine learning models to optimize cache hit rates, further enhancing privacy and system performance. We will also address real-world challenges in IoV environments, considering factors like network variability and vehicle mobility. By refining the scheme based on these practical issues, we aim to ensure its robustness and scalability in diverse IoV scenarios.

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